

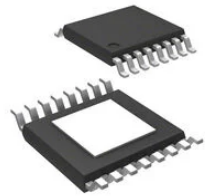
Motor&Wheels

Option	Description	Pros	Cons	Device
1 \$9.69	4 Sets TT Motor with Leads, DC 3-6V Gearbox Motor Dual Shaft 200RPM Ratio 1:48 Motor with Tire Wheel Kit	- Meets size requirement - Lower-end price 2-in-1 wheel and motor kit	Lower voltage means less power to wheels	 Motor with dual wheel assembly
1 \$16.50	GEARMOTOR 251 RPM 12V W/ENCODER	Meets force requirement	- High Amp-cap - Only powers 1 wheel	 1738-1106-ND
3 \$27.45	GEARMOTOR 85 RPM 12V MICRO METAL	- Meets force requirement - Mid range Amp	- Mid price range - Only power 1 wheel	 21183-4799-ND

Choice: Option 1

Rationale: Option 3 was the motor used last semester and had the most promise as it was proven to be dependable. Ultimately option 1 was chosen due to its price, addition of compatible wheel components and quantity.

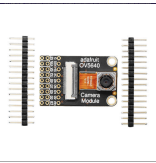
Motor Driver

Option	Description	Pros	Cons	Device
1 \$3.02	Bipolar Motor Driver NMOS PWM 16-HTSSOP	- Meets Voltage and Amp requirements and - Available in stock	Potentially difficult to program	 DRV8833PWP R
2 \$1.49	IC HALF BRIDGE DRVR 250MA 14SOIC	Meets Voltage and AMP requirement	- Not in Stock - Not much higher Amp supply than motor demands	 LB1837M
3 Obsolete	IC MOTOR DRIVER 2.5V-7.5V 16DMP	Meets Voltage requirements	- No longer made - No amp rating	 NJM2611M

Choice: Option 1

Rationale: Option 1 was the best option due to its availability. It also meets the requirements when dealing with both the Voltage range and the Amp output for the motors that it will be controlling

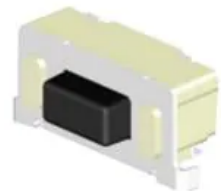
light plus camera

Option	Description	Pros	Cons	Device
1 \$14.95	OV5640 Camera Sensor Platform Evaluation Expansion Board	- Meets Voltage and Amp requirements and - Available in stock	Potentially difficult to program	 https://shorturl.at/JOr7s
2 \$56.04	ARDUCAM 1080P DAY/NIGHT VISION U	- Has built in camera and light module - Meet power specs	Doesn't meet the surface mounting requirement	 https://shorturl.at/83nms
3 \$30.00	REALTEK AMB82 MINI IOT AI CAMERA	In stock	- Doesn't require surface mounting - Cost too much	 https://shorturl.at/ARJbj

Choice: Option 1

Rationale: Option 1 was the best option due to its availability. It also meets the requirements when dealing with both the Voltage range as well as being compatible with the led set up were using.

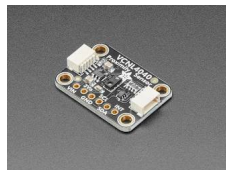
Push Button

Option	Description	Pros	Cons	Device
1 \$0.59	TL1105JAF100Q SWITCH TACTILE SPST-NO 0.05A 12V	Requires only 100gf operating force	Takes up more vertical space	 https://shorturl.at/T89WN
2 \$0.42	PTS645VK31-2 LFS SWITCH TACTILE SPST-NO 0.05A 12V	gold/silver dome contacts provide longer life with esp32	Requires ~160gf to push	 https://shorturl.at/sLBsE
3 \$0.82	1188EP-1K3-V-T/R Tactile Switches Side Push Type 3.5*7	- Surface mounted - Minimal clearance impact, short and compact	- Lacks through-hole anchor pins - Tight soldering tolerances	 https://shorturl.at/1jPEq

Choice: Option 2

Rationale: Option 2 is the lowest cost option in the table while maintaining long contact life via its gold/silver dome construction. It also uses through-hole legs that can withstand repeated right-angle horizontal force without the risk of shearing off PCB surface pads.

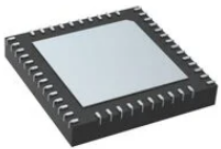
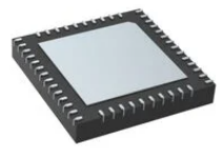
Position Sensor

Option	Description	Pros	Cons	Device
1 \$10.86	GP2Y0A21YK0F SENSOR OPTICAL ANALOG	- Simple 0–3V linear-ish voltage output - provides consistent range measurements	- Needs voltage step up boost converter - Large frame/footprint required	 https://shorturl.at/mdYmp
2 \$9.57	ST0242 VL53L0X DISTANCE SENSOR	- Operates down to 2.6V–3.3V - Compact footprint - precise millimeter-level distance measurements	- Requires careful optical window design - requires more firmware initialization	 https://shorturl.at/okf9M
3 \$5.95	4161 VCNL4040 PROXIMITY AND LUX SENSORS	- Fully functional on a 2.5V–3.6V rail - Compact footprint	- Limited to short-range proximity (~200 mm max) - Reflective material dependency	 https://www.digkey.com/en/products/detail/adafruit-industries-llc/4161/10385108

Choice: Option 2

Rationale: Option 2 is the best choice because it operates within the regulated 3V power rail without requiring an extra voltage step-up converter. Additionally, its laser Time-of-Flight technology provides precise, millimeter-level distance measurements that are not limited by short range or surface reflectivity variations.

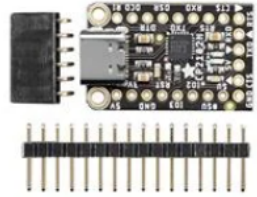
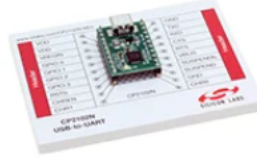
ESP 32

Option	Description	Pros	Cons	Device
1 \$1.84	ESP32-D0W D-V3	• Low cost • Dual-core • Wi-Fi + Bluetooth	• More complicated PCB • Older-generation	 https://www.digikkey.com/en/products/detail/espressif-systems/ESP32-D0WD-V3/11657616
2 \$3.00	ESP32-D0W D	• Same General architecture as ESP32 • Large community support	• Digi-Key marks it not recommended • Older silicon	 https://www.digikkey.com/en/products/detail/espressif-systems/ESP32-D0WD/8028403
3 \$3.00	ESP32-D0W DQ6	• Physically larger • Lots of documentation	• 6x6 instead of 5x5 • Difficult to assemble by hand	 https://www.digikkey.com/en/products/detail/espressif-systems/ESP32-D0WDQ6/8028411

Choice: Option 1

Rationale: It's small so it takes up less space, and it's one of the newer versions so it's just as good as the others even though it's cheaper. It also has Wi-Fi plus Bluetooth capabilities In case we end up needing it

ESP 32 Programmer

Option	Description	Pros	Cons	Device
1 \$5.95	Adafruit 5335 CP2102N	• Cheap • Easy to use • Small • Good documentation	• Mostly a programmer not debugger • Requires self made connections	 https://www.digikey.com/en/products/detail/adafruit-industries-llc/5335/15913267
2 \$13.86	Silicon Labs CP2102N-MINIEK	• Reliable • Good manufacturer support	• Expensive • Overkill • Larger than needed	 https://www.digikey.com/en/products/detail/silicon-labs/CP2102N-MINIEK/6605110
3 \$8.90	Seeed 103990049 CH340	• Lots of support • Good availability	• Less polished • No made for esp32 specifically	 https://www.digikey.com/en/products/detail/seeed-technology-co-ltd/103990049/5487819

Choice: Option 1

Rationale: it's compact and cheap while having 3.3 V Logic and simple to use once I make some connectors. It doesn't require many connections and has good documentation